

β-glucan reprograms alveolar macrophages via neutrophil/IFNγ axis to promote lung injury

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eLife Assessment

This **important** study advances our understanding of maladaptive innate immune training. The experimental evidence supporting the conclusions is **convincing** and the expert reviewers strongly endorse the manuscript. The work will be of high interest to both researchers in the trained immunity field and clinician scientists.

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Abstract

Alveolar macrophages (AMs) reside in the lower airways and play a crucial role in lung health and response to sterile inflammation and infections. AMs possess remarkable adaptability to different environmental challenges that can persist through their memory capacity (trained immunity). β-glucan has been characterized as a potent inducer of central trained immunity by reprogramming hematopoietic stem cells (HSCs) in the bone marrow. In the present study, we show that systemic administration of β-glucan induces peripheral trained immunity by reprogramming alveolar macrophages (AMs) in the lungs, in a Dectin1-independent manner. We furthermore demonstrate that AM reprogramming at both the transcriptional and metabolic levels exacerbate lung injury following bacterial (LPS) or viral (polyI:C) challenges via a neutrophil/IFN-γ dependent manner. These findings identify an additional facet of β-glucan in trained immunity involving AM reprogramming and shed light on the potential detrimental effects of trained immunity.

Introduction

The lungs represent a significant infective niche for numerous infectious agents. However, severe pulmonary disease is increasingly being associated with immune-mediated responses rather than direct damage of invading microbes^{1,2}. Among the various triggers of immunopathology, lipopolysaccharide (LPS), a component of the outer membrane of Gram-negative bacteria, is a potent inducer of systemic inflammation via activation of Toll-like receptor 4 (TLR4)³. Excessive inflammation associated with LPS can cause acute lung injury (ALI), a severe inflammatory condition of the lungs characterized by alveolar damage, edema, and impaired gas exchange leading to organ dysfunction, and ultimately, mortality^{4,5}. Thus, the magnitude of inflammatory responses must be regulated to resolve infections while preventing collateral tissue damage. Understanding the mechanisms underlying the regulation of inflammation in ALI is essential for the development of effective therapeutic interventions. Key players in this regulatory network include various immune cells, cytokines, and signalling pathways, which collectively modulate the intensity and duration of inflammatory responses. Dysregulation of these mechanisms can lead to either persistent inflammation or immunosuppression, both of which contribute to the pathogenesis of sepsis^{6,7,8}.

The constant exposure of the lungs to a non-sterile environment leads to the development of a unique immunity by eliminating inhaled foreign particles while minimizing inflammatory-mediated tissue damage. Resident alveolar macrophages (AMs) serve a crucial regulatory function to maintain the delicate balance between inflammation and protecting lung tissue from damage⁹. AMs originate from fetal liver monocytes and act as vigilant sentinels protecting the airways against invading pathogens and pollutants^{10,11}. Thus, AMs play a pivotal role in orchestrating both the initiation and resolution of immune responses within the lung microenvironment.

Adaptations in innate immune cells are diverse with substantial plasticity to various insults that can be maintained resulting in enhanced (trained immunity) or reduced (tolerance) inflammatory responses to a second stimuli^{12,13,14}. Trained immunity is mediated *via* long-term metabolic reprogramming and epigenetic modifications, which can be induced by various stimuli such as the attenuated mycobacteria *Bacille Calmette-Guérin* (BCG) or β -glucan (a polysaccharidic component of fungi cell wall)^{12,15}. BCG and β -glucan are able to train monocytes and neutrophils with beneficial impact in cancer or infections *via* the reprogramming of haematopoietic stem cells (HSC) within the bone marrow^{16,17,18,19,20}. However, there are substantial knowledge gaps in our understanding of trained immunity within tissue-resident macrophages (TRMs) and its regulatory role in maintaining tissue homeostasis under stress conditions. For instance, while the initial LPS-stimulation of macrophages induces a strong inflammatory responses *via* TLR-4, restimulation with LPS generates tolerance in these macrophages²¹. Thus, the epigenetic reprogramming of innate immune cells and their subsequent response depends on two signals: the initial training agent and the nature of the second stimuli at an inflammatory site. Here we showed that systemic administration of β -glucan can reprogram AMs in the lungs *via* neutrophil/type II IFN axis but independent of Dectin-1 signalling. The unique transcriptomic and metabolic profile of AMs render them hyperresponsive to both bacterial or viral stimulation causing dysregulated inflammatory responses with pulmonary damage. The differences in the systemic basal levels of β -glucan in sepsis patients as well as the pulmonary levels of IFN γ can be major factors in determining the hyperresponsiveness in sepsis patients and can be potentially targeted for therapy.

Results

β -glucan-mediated trained immunity aggravates LPS-induced ALI

To investigate whether systemic administration of β -glucan induces trained immunity within the lung, we sought to assess its impact on Acute Lung Injury (ALI) triggered by LPS treatment. We initially evaluated the consequences of β -glucan-mediated training on ALI by administering LPS seven days after training (**Figure 1A**). ALI assessment conducted 24 hours post-LPS instillation did not reveal any discernible distinctions attributable to the initial i.p. β -glucan injection. However, ALI was notably exacerbated in mice that had undergone β -glucan training upon subsequent LPS stimulation as demonstrated non-invasively by lung microCT scanners with regards to the increased proportion of poorly- or non-aerated lung segment and to a significant increase in Hounsfield units (**Figure 1B**). Hounsfield units reflect a coefficient of tissue attenuation and, by extrapolation, tissue composition. Consistent findings were corroborated through two complementary invasive evaluations of ALI. Specifically, the heightened alveolar-capillary permeability following LPS instillation in β -glucan-trained mice was substantiated by elevated lung Evans blue dye concentrations (**Figure 1C**). Histological examination further unveiled a spectrum of notable alterations following LPS instillation, including cellular infiltration, thickened alveolar walls, and the formation of hyaline membranes, observed in β -glucan-trained mice and not PBS controls (**Figure 1D**). We additionally monitored weight loss following LPS instillation (**Supplementary Figure 1A**). ALI-induced inflammation peaks at 24h, reflected in the weight loss of the mice which drops nearly 10% within a day. While there were no differences in weight loss between control and β -glucan-treated mice at this timepoint, we observed that β -glucan mice exhibited a delayed recovery period given they did not return to their original weight by day 4, as opposed to control mice (**Supplementary Figure 1B**).

The increased cell infiltration 24 hours after LPS instillation in β -glucan-treated mice was due to a significant recruitment of neutrophils, as assessed in the bronchoalveolar lavage (BAL; CD11b⁺Ly6G⁺). Strikingly, the frequency and total numbers of neutrophils was doubled in the β -glucan-treated mice, after LPS administration (**Figure 1E**). Concomitant with the increased neutrophil recruitment, CXCL1 was significantly increased in β -glucan-treated mice when compared to control mice after LPS treatment (**Figure 1F**).²² Amplified ALI in β -glucan-trained mice was also associated with increased proinflammatory cytokines (IL-6 and TNF α) (**Figure 1G-H**). As AMs are major producers of CXCL1, IL6 and TNF α , we next assessed whether β -glucan increased the proportion or number of AMs and found no significant differences between naïve and β -glucan trained mice (**Figure 1I**). There was also no changes in the frequency of CD11b⁺ AMs at baseline and post-LPS instillation between β -glucan-treated and control mice (**Supplementary Figure 1C**). This suggests that β -glucan does not induce the expansion of monocyte-derived AMs. To further characterize AM phenotype following β -glucan, we assessed several AM-associated cell surface markers and found no differences in expression of MHCII, F4/80, CD169 and CD69 (**Supplementary Figure 1D**). Interestingly, β -glucan enhanced the frequency of CD80⁺ AMs, a costimulatory molecule associated with activation and pro-inflammatory responses (**Supplementary Figure 1D**).²³ Thus the increased cytokine production was not due to an increased number of AMs, but rather intrinsic functional changes.

Next, we examined whether the effects of β -glucan-induced immune training persisted in ALI severity. To do so, we administered β -glucan or PBS (control) in mice and after 28 days challenged them with LPS (**Supplementary Figure 1E**). Similar to the 7-day timeframe of β -glucan training, there was increased endothelial permeability of the lungs (**Supplementary Figure 1F**), and increased immunopathology (**Supplementary Figure 1G**) in 28-days β -glucan trained mice challenged with LPS. This increased ALI in β -glucan treated mice after 28 days was associated with a higher concentration of CXCL1, IL6 and TNF α (**Supplementary Figure 1H**) and increased

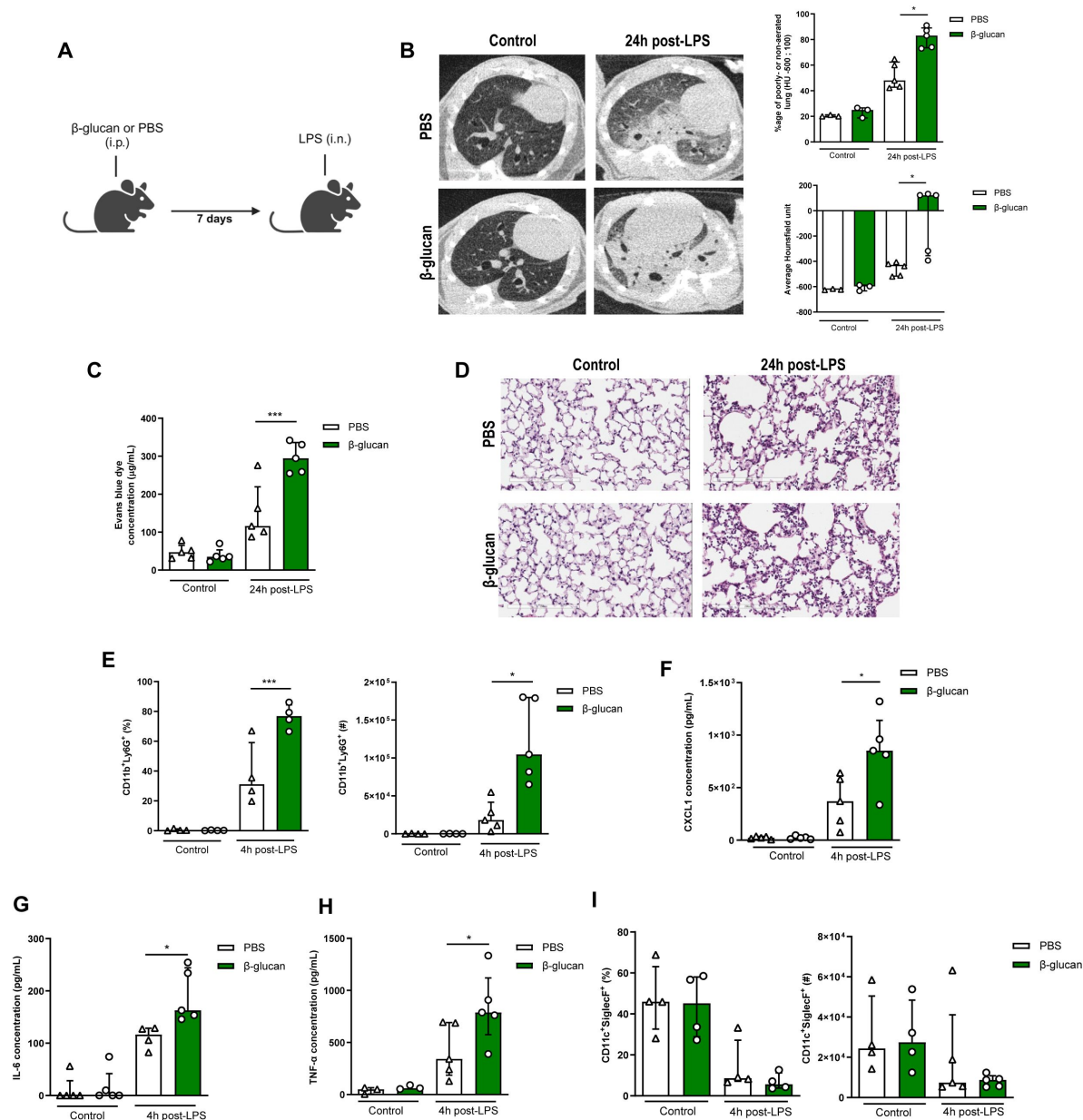


Figure 1.

β -glucan-mediated trained immunity increases LPS-induced ALI.

A, Schematic of the β -glucan-induced training seven days before lipopolysaccharide (LPS)-induced acute lung injury (ALI) model. Experiments were performed in sex- and age-matched 10-12 weeks old control (i.p. PBS, white bars) and trained (i.p. β -glucan, green bars) WT mice. **B**, Lung micro-CT scan, percentage of poorly- or non-aerated lung and average lung Hounsfield unit. **C**, Alveolar capillary membrane permeability assessed by lung Evans blue dye concentration. **D**, Lung histology after staining with haematoxylin and eosin. **E**, Quantification of BAL neutrophils frequency (left) and absolute count (right) (gated on single live cells, CD45.2⁺, CD11c⁻, Siglec-F⁻, CD11b⁺, Ly6G⁺). **F**, **G**, **H** BAL chemokine and pro-inflammatory cytokines concentrations (left to right) (CXCL1: chemokine C-X-C motif ligand 1, IL-6: interleukin-6 and TNF- α : tumor necrosis factor α). **I**, Quantification of BAL alveolar macrophages frequency (left) and absolute count (right) (gated on single live cells, CD45.2⁺, CD11c⁺, Siglec-F⁺). Data were analysed using one-way ANOVA followed by Dunn's multiple comparisons test. * $p < 0.05$, *** $p < 0.001$.

neutrophil infiltration in BAL (**Supplementary Figure 1I** [↗](#)) following LPS instillation. The frequency and number of AMs were not changed between groups (**Supplementary Figure 1J** [↗](#)). Increased production of cytokines was also observed upon *ex vivo* stimulation of AMs isolated from treated mice 28-day post- β -glucan (**Supplementary Figure 1K, L** [↗](#)). Thus, the intrinsic functional changes in AMs suggest that the exacerbation of ALI induced by β -glucan triggered a long-term reprogramming of immune cells rather than an additive effect of lingering inflammation from β -glucan injection. To assess whether this response was specific to bacterial LPS or viral agonists can cause similar responses, we next challenged β -glucan trained mice with a TLR-3 agonist (poly(I:C)) (**Supplementary Figure 2A** [↗](#)). Similar to the LPS-model, β -glucan treated mice had heightened poly(I:C)-induced ALI shown via increased alveolar-capillary permeability, tissue damage (**Supplementary Figure 2A-C** [↗](#)). Pro-inflammatory cytokines CXCL1 and IL-6 were also increased while TNF α remains unchanged following challenge suggesting differing cytokine profiles between LPS and Poly(I:C)-induced ALI (**Supplementary Figure 2D** [↗](#)). However, β -glucan-treated mice similarly had increased neutrophil infiltration following Poly(I:C) instillation (**Supplementary Figure 2E** [↗](#)). The number of AMs was also unchanged among groups (**Supplementary Figure 2F** [↗](#)). Collectively, these results suggest that systemic β -glucan can maintain a long-term reprogramming in AMs promoting ALI.

β -glucan augmented ALI is mediated via AM

Although there were no differences in the frequency or absolute number of AMs between β -glucan-treated and control mice, the early heightened response with increased CXCL1 production and the recruitment of neutrophils indicate that the AMs are engaged in the exacerbation of ALI induced by β -glucan. To further characterize the role of these cells in ALI, we locally depleted AMs using intranasally administered clodronate liposomes 2 days before performing the LPS instillation, which is at its peak of AM depletion (**Figure 2A** [↗](#)). Depletion of AMs significantly reduced the production of cytokines TNF α , IL-6, and CXCL1 and the recruitment of neutrophils in BAL, which abolished the β -glucan-induced ALI (**Figure 2B-D** [↗](#)). To further confirm that tissue-resident AMs are responsible for the increased β -glucan-induced ALI, we used *Csf2rb*^{-/-} mice (**Figure 2E** [↗](#)), which do not naturally develop alveolar macrophages throughout their lifespan but maintain regular levels of bone marrow derived macrophages (BMDMs) and interstitial macrophages²⁴ [↗](#). Similar to the AM depletion, the production of cytokines and neutrophil recruitment in BAL was similar between β -glucan treated and control *Csf2rb*^{-/-} mice after LPS administration. Importantly, there was no difference in ALI, as similar levels of proteins were measured in the BAL of β -glucan treated and control *Csf2rb*^{-/-} after the LPS challenge (**Figure 2F-H** [↗](#)).

To complement these loss-of-function experiments with a gain-of-function, we adoptively transferred AMs from adult WT control (PBS) or trained (β -glucan) mice to day 2 *Csf2rb*^{-/-} neonate mice (**Figure 2I** [↗](#)) and after 6 weeks challenged these mice with LPS. Remarkably, mice replenished with AMs from β -glucan-trained mice displayed an increased in production of inflammatory cytokines and the recruitment of neutrophils into the BAL (**Figure 2K** [↗](#), L). This was associated with increased ALI as they had higher protein levels in their BAL compared to mice who received control AMs (**Figure 2J** [↗](#)). Collectively, our data demonstrate that the exacerbation of LPS-induced ALI in β -glucan-trained mice is mediated through AMs, mainly via functional reprogramming rather than a change in the proportion or number of AMs.

β -glucan reprograms AM

Our findings indicate that β -glucan reprograms AMs causing an increased response to LPS and subsequently enhanced ALI. To test this hypothesis, we investigated the transcriptional state of AMs after seven days training with β -glucan *in vivo*, which were then stimulated with/without LPS *ex vivo* LPS. The rationale for using an *ex vivo* system was to ensure that AMs are uniformly stimulated with LPS (**Figure 3A** [↗](#)).

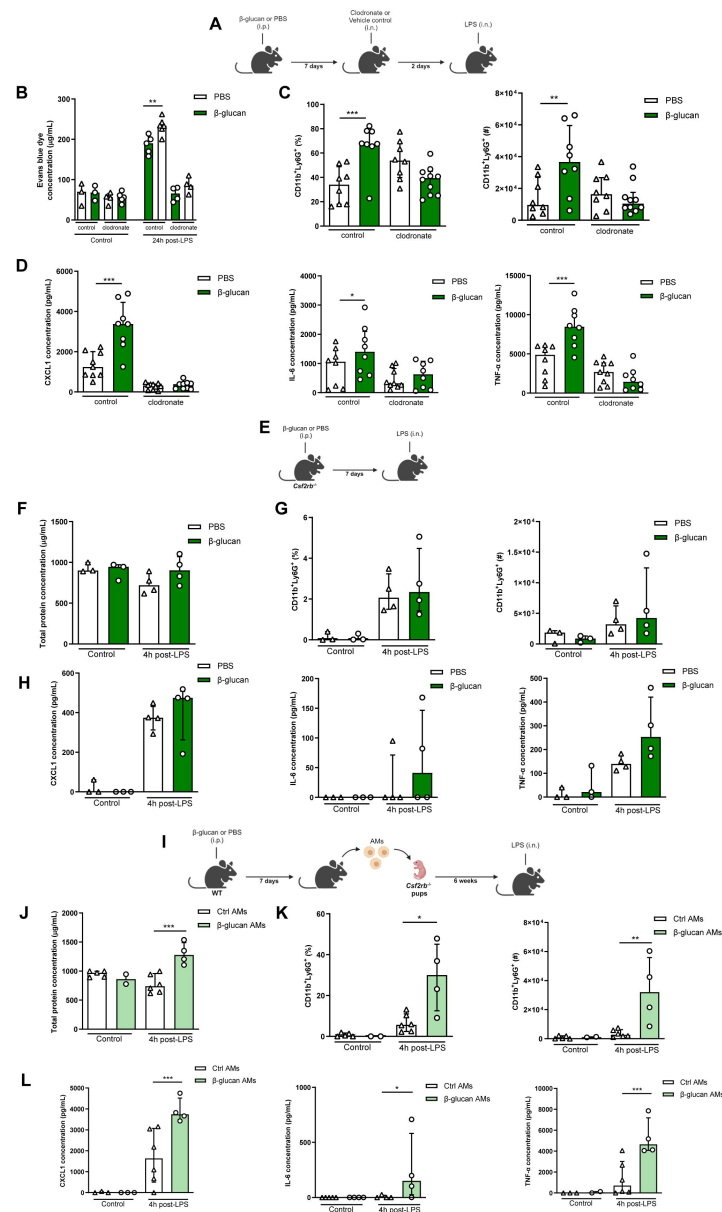


Figure 2.

Systemic administration of β -glucan enhances ALI via AMs.

A, Schematic of the clodronate-mediated alveolar macrophages (AM) depletion experiments, performed in sex- and age-matched 10-12 weeks old control (i.p. PBS, white bars) and trained (i. p. β -glucan, green bars) WT mice. **B**, Alveolar capillary membrane permeability assessed by lung Evans blue dye concentration. **C**, Quantification of BAL neutrophils frequency (left) and absolute count (right) (gated on single live cells, CD45.2⁺, CD11c⁻, Siglec-F⁺, CD11b⁺, Ly6G⁺). **D**, BAL chemokine C-X-C motif ligand 1 (CXCL1) and pro-inflammatory cytokines (IL-6: interleukin-6 and TNF- α : tumor necrosis factor α) concentrations. **E**, Schematic of the β -glucan-induced training and lipopolysaccharide (LPS)-induced acute lung injury (ALI) model in sex- and age-matched 6 weeks old control (i.p. PBS, white bars) and trained (i. p. β -glucan, green bars) *Csf2rb*^{-/-} mice. **F**, BAL total protein concentration. **G**, Quantification of BAL neutrophils frequency (left) and absolute count (right) (gated on single live cells, CD45.2⁺, CD11c⁻, Siglec-F⁺, CD11b⁺, Ly6G⁺). **H**, BAL CXCL1, IL-6 and TNF- α concentrations. **I**, Schematic of the adoptive transfer of control (i.p. PBS, white bars) or β -glucan-trained (i. p. β -glucan, green bars) AMs collected from adult WT mice to 2 days old *Csf2rb*^{-/-} mice. Lipopolysaccharide (LPS)-induced acute lung injury (ALI) was performed 6 weeks after adoptive transfer. **J**, BAL total protein concentration. **K**, Quantification of BAL neutrophils frequency (left) and absolute count (right) (gated on single live cells, CD45.2⁺, CD11c⁻, Siglec-F⁺, CD11b⁺, Ly6G⁺). **L**, BAL CXCL1, IL-6 and TNF- α concentrations. Data were analysed using one-way ANOVA followed by Dunn's multiple comparisons test. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

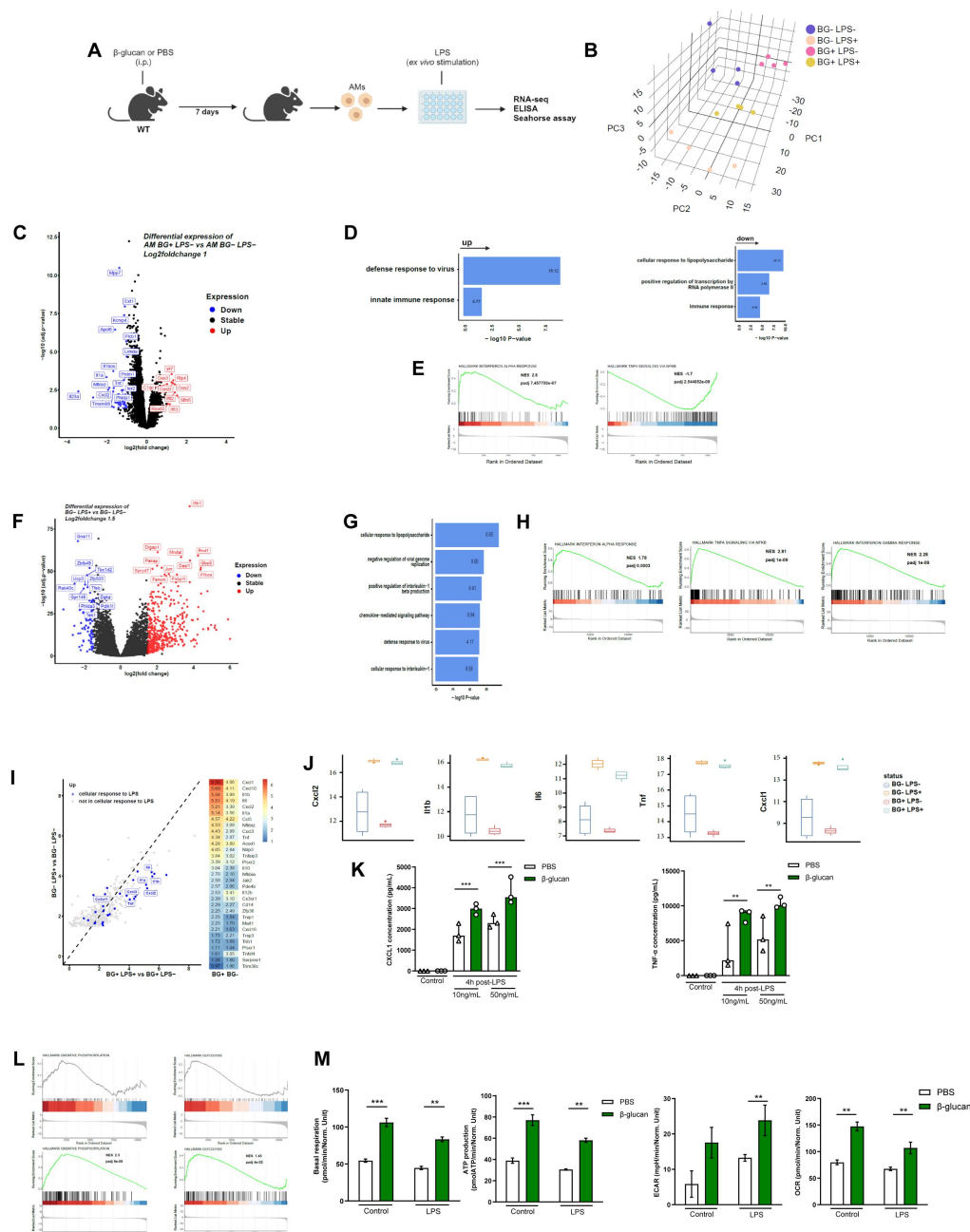


Figure 3.

β-glucan reprograms AMs.

A, Schematic of control (i.p. PBS) or β-glucan-trained (i.p. β-glucan) AMs collected from adult WT mice *ex vivo* stimulation with LPS (LPS-: unstimulated, LPS+: stimulated in RNAseq analysis). **B**, Discovery plot. **C**, AM differential expression of genes in response to β-glucan training. **D**, Gene ontology in response to β-glucan training. **E**, GSEA in response to β-glucan training. **F**, AM differential expression of genes in response to LPS stimulation. **G**, Gene ontology in response to LPS stimulation. **H**, GSEA in response to LPS stimulation. **I**, AM gene expression in response to LPS in β-glucan-trained AMs. **J**, Examples of genes expression in response to LPS in control vs β-glucan-trained AMs. **K**, Chemokine C-X-C motif ligand 1 (CXCL1) and tumor necrosis factor α (TNF-α) concentrations after *ex vivo* LPS stimulation. **L**, GSEA of oxidative phosphorylation (left) and glycolysis (right) pathways according to β-glucan-training in unstimulated (LPS-) and LPS stimulated (LPS+) AMs. **M**, Evaluation of AM metabolism: basal respiration (upper left), ATP production (upper right), extracellular acidification rate (ECAR, lower left), oxygen consumption rate (OCR, lower right). Data were analysed using one-way ANOVA followed by Dunn's multiple comparisons test. ** $p < 0.01$, *** $p < 0.001$.

Principal component analysis (PCA) of all expressed genes across the four conditions revealed that control and β -glucan trained AMs had distinct transcriptional profiles (**Figure 3B**). More precisely, the 4 different conditions explained the largest variance as they spread out across the first principal component (PC), while the variation within replicates was associated to the second and third PC. Next, we investigated the effect of β -glucan trained AMs by comparing the differential gene expression between control and β -glucan treated mice (**Figure 3C**). We found 88 differentially expressed genes (32 up-regulated and 56 down-regulated, with a false discovery rate (FDR) adjusted p value < 0.05 and a log2fold-change cut-off of 1). Despite that only few genes were differentially expressed, gene ontology analysis indicated a strong enrichment of genes involved in innate immune response and defense response to virus (**Figure 3D**). Interestingly, LPS response genes were downregulated in β -glucan-treated mice suggesting that, in response to β -glucan, the LPS response pathway is attenuated or less activated. Gene set enrichment analysis (GSEA) additionally revealed an increase of genes involved in IFN α but a significant decrease of genes involved in TNF α signalling (**Figure 3E**). Next, we explored how *ex vivo* stimulated AMs responded to LPS using the same differential gene expression analysis. LPS altered the expression of 525 genes (FDR > 0.05 log2FC > 1.5) with the up regulation of 438 genes including numerous pro-inflammatory genes (**Figure 3F**). As expected, gene ontology analyses revealed that these genes were involved in cellular response to LPS, inhibition of viral genome replication, production of interleukin-1 β and cellular response to IL-1 (**Figure 3G**). Additional analysis of GSEA indicated up-regulation of genes involved in TNF α signalling (**Figure 3H**), indicating that LPS and β -glucan both induce interferon responses but have opposite effects on the TNF response pathway.

To delineate how β -glucan training of AMs may affect the subsequent response to LPS, we then compared the fold change of gene expression in response to LPS stimulation between β -glucan trained and untrained AMs. Notably, we observed an upregulation of genes associated with the LPS response (depicted in blue) in the β -glucan trained group (**Figure 3I**). This heightened response included a significant increase in the expression of IL-6, IL-1, TNF, and other genes (**Figure 3I, J**). This increased response to LPS by β -glucan-trained AMs was confirmed at the protein level as they exhibited increased *ex vivo* production of CXCL1 and TNF α (**Figure 3K**). Such an increased response to LPS was also obtained in AMs after 28 days β -glucan training (**Supplementary Figure 1G**). We also observed similar trends in genes associated with the defense response to viruses (**Supplementary Figure 2G**) which is in line with our findings showing an increased ALI to poly(I:C) (**Supplementary Figure 2B-D**). Taken together, β -glucan reprograms AMs to respond robustly to LPS stimulation.

As trained immunity is associated with metabolic rewiring, we next assessed if β -glucan-induced training modified the AM's metabolic state. Genes involved in both oxidative phosphorylation and glycolysis were up-regulated during response to LPS in β -glucan-trained AMs compared to untrained AMs but not at the steady state (**Figure 3L**). The Seahorse assay showed while only mitochondrial respiration was increased in β -glucan-trained AMs at the steady state, both mitochondrial respiration and glycolysis were increased during the response to LPS in β -glucan-trained AMs, (**Figure 3M**). Taken together, β -glucan functionally reprograms AMs by rewiring their transcriptomic and metabolic states.

IFN γ signaling and neutrophils are required for AM reprogramming

Dectin-1 is the receptor that recognizes β -glucan and is involved in mediating the biological effects of β -glucan-induced training in most cell types¹⁵. To assess the role of Dectin-1 in β -glucan increased LPS-induced ALI, we used the same experimental setup using Dectin-1 deficient mice (**Supplementary Figure 3A**). Surprisingly, the increased neutrophil infiltration and pro-inflammatory cytokine production in BAL were maintained in β -glucan-treated *Dectin-1*^{-/-} mice and there was no change in the AM numbers (**Supplementary Figure 3B-D**). To determine the effect of Dectin-1 on β -glucan-induced AM function, *Dectin-1*^{-/-} mice were treated with β -glucan

and after 7 days AMs were cultured and stimulated *ex vivo* with LPS (**Supplementary Figure 3E**). Similar to wild-type AMs, the production of pro-inflammatory cytokines CXCL1 and TNF α was increased in β -glucan-trained Dectin1-deficient AMs after LPS stimulation (**Supplementary Figure 3F**). This indicates that β -glucan-induced AMs reprogramming is Dectin-1-independent and that AMs are trained via a different signalling pathway. Since we observed type I IFN gene expression is increased in AMs after β -glucan treatment (**Figure 3E**) and type I IFN was previously reported to be involved in AMs training²⁵, we next assessed the impact of β -glucan on AMs reprogramming in *Ifnar*^{-/-} mice, which lack type I IFN signalling (**Supplementary Figure 3G**). Similar to Dectin-1 deficient mice, increased production of pro-inflammatory cytokines in response to LPS by β -glucan-treated *Ifnar*^{-/-} AMs was maintained, demonstrating that β -glucan-induced AMs reprogramming is type I IFN-independent (**Supplementary Figure 3H**).

Considering recent studies suggesting type II IFN can train AMs following BCG²⁶ or adenoviral infection²⁷, using *Ifng*^{-/-} mice we next investigated if type II IFN is involved in β -glucan-induced exacerbation of ALI (**Figure 4A**). In contrast to *Dectin1*^{-/-} or *Ifnar*^{-/-} mice, the increased neutrophils recruitment, inflammatory cytokines production, and ALI were abolished in β -glucan-trained *Ifng*^{-/-} mice (**Figure 4B-D**). To investigate the role of type II IFN signalling in β -glucan-induced reprogramming of AMs, we performed *ex vivo* LPS stimulation on β -glucan trained *Ifng*^{-/-} AMs (**Figure 4E**). The increase of inflammatory cytokines in response to LPS by β -glucan-trained AMs was abolished (**Figure 4F**), suggesting that training of AM by β -glucan is IFN γ -dependent.

Considering the concentration of IFN γ was significantly increased in BAL at day 1 post- β -glucan administration, which was associated with the increased number of neutrophils (**Figure 4G-J**), we next assess the direct role of IFN γ in training of AMs by β -glucan. We selectively depleted IFN γ prior to β -glucan administration *in vivo* (**Figure 4K**), and found the production of inflammatory cytokines (CXCL1 and TNF α) was significantly reduced in β -glucan trained AMs. Similarly, the depletion of neutrophils prior to β -glucan treatment in mice resulted in significant reduction in inflammatory cytokines in β -glucan trained AMs (**Figure 4L**). These findings indicate that β -glucan reprograms AMs via IFN γ - and neutrophils-dependent manner.

Discussion

The evolving field of trained immunity has challenged the boundaries we once drew to discern between the innate and adaptive immune systems. The discovery that the innate immune system can be ‘trained’ to retain memory-like features has provided novel avenues for prophylactic and therapeutic strategies. The use of an adjuvant like β -glucan has been investigated as an anti-infection agent in acute infections or inflammatory conditions^{16, 28} as well as an anti-cancer treatment^{29, 30}. While we and others have shown that β -glucan can induce central trained immunity by functionally reprogramming the hematopoietic stem cell compartment in the bone marrow, the present study highlights an additional facet of β -glucan’s immunomodulatory effects on residential immune cells. Here we show that a single intraperitoneal (i.p.) injection of β -glucan can induce trained immunity in the lung alveolar macrophages and functionally reprograms AMs at both the transcriptional and metabolic levels. These findings underscore the occurrence of tissue-specific immune training following systemic treatment, demonstrating processes of β -glucan-mediated peripheral trained immunity occurring in parallel with central immunity. We subsequently demonstrate that β -glucan trained AMs can be detrimental after LPS or poly(I:C) administration causing severe ALI. Thus, this study is shedding light on the deleterious effects of trained immunity and its impact on immunopathology, which is incompletely understood.

There are several key features in trained immunity, including shifts in epigenetic and transcriptomic profiles after initial stimulation that returns to steady states. However, this epigenetic and metabolic remodeling result in long-term memory with heightened responses to diverse secondary stimulation¹². The nature of each stimulus appears to be critical in

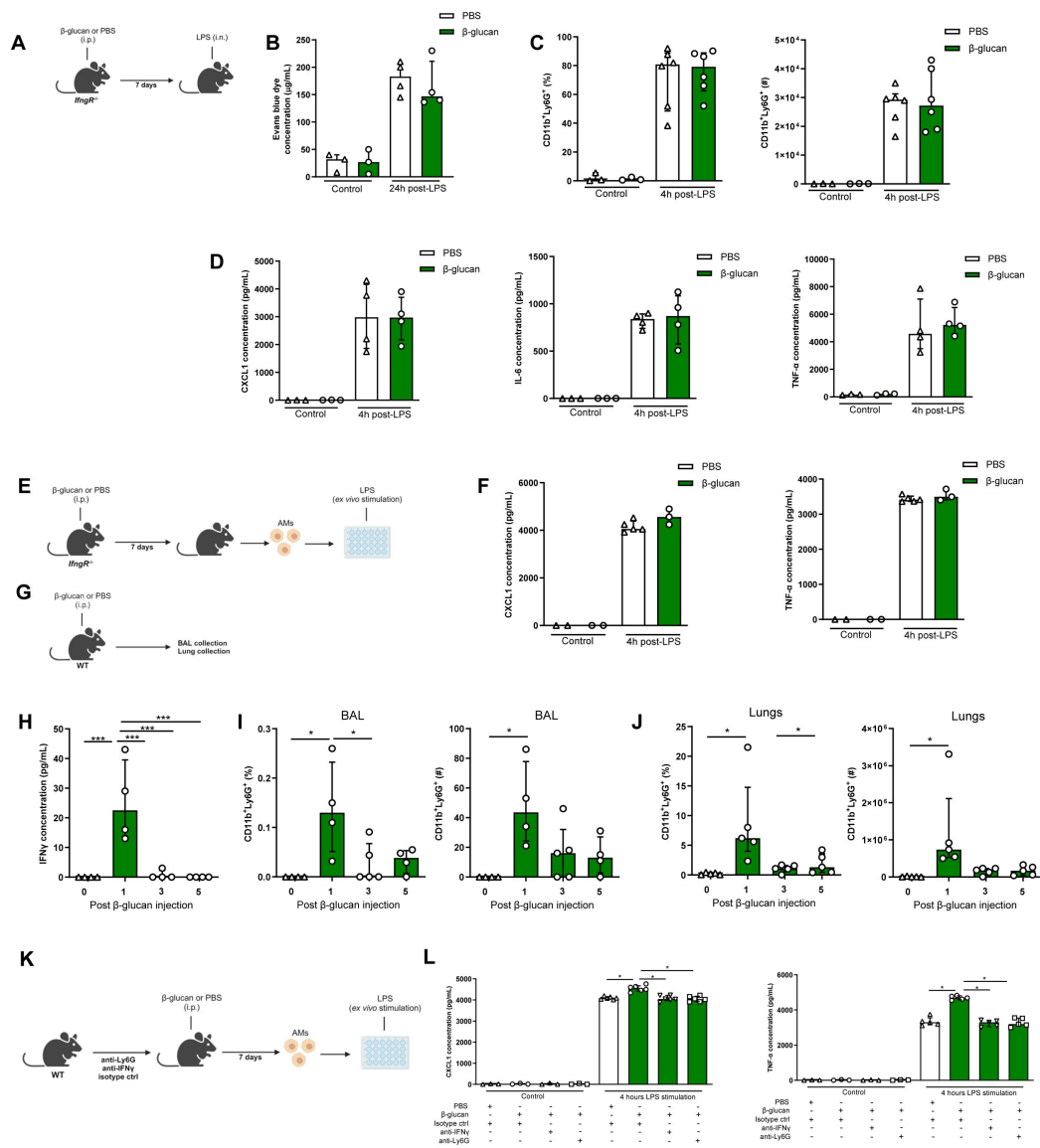


Figure 4.

IFN γ and neutrophils are required in β -glucan-mediated AM reprogramming.

A, Schematic of the β -glucan-induced training and lipopolysaccharide (LPS)-induced acute lung injury (ALI) model. Experiments were performed in sex- and age-matched 10-12 weeks old control (i.p. PBS, white bars) and trained (i.p. β -glucan, green bars) *IfngR*^{-/-} mice. **B**, Alveolar capillary membrane permeability assessed by lung Evans blue dye concentration. **C**, Quantification of BAL neutrophils proportion (left) and absolute count (right) (gated on single live cells, CD45.2⁺, CD11c⁻, Siglec-F⁻, CD11b⁺, Ly6G⁺). **D**, BAL chemokine C-X-C motif ligand 1 (CXCL1) concentration (left) and pro-inflammatory cytokines (IL-6: interleukin-6 – middle- and TNF- α : tumor necrosis factor α – right) concentrations. **E**, Schematic of control (i.p. PBS, white bars) or β -glucan-trained (i.p. β -glucan, green bars) AMs collected from adult *IfngR*^{-/-} mice ex vivo stimulation with LPS. **F**, Chemokine C-X-C motif ligand 1 (CXCL1) and tumor necrosis factor α (TNF- α) concentrations after ex vivo LPS stimulation. **G**, Schematic of the analysis of the effect of i.p. β -glucan injection on interferon- γ (IFN γ) production and neutrophils expansion before (white bars) and Day 1, Day 3, Day 5 and Day 7 post-injection (green bars). **H**, BAL IFN γ concentrations. **I**, Quantification of BAL neutrophils proportion (left) and absolute count (right) (gated on single live cells, CD45.2⁺, CD11c⁻, Siglec-F⁻, CD11b⁺, Ly6G⁺). **J**, Quantification of lung neutrophils proportion (left) and absolute count (right). **K**, Schematic of control (i.p. PBS, white bars) or β -glucan-trained (i.p. β -glucan, green bars) AMs ex vivo stimulation with 50ng/mL LPS. AMs were collected from control (i.p. injection of isotypes), neutrophils depleted (i.p. injection of anti-Ly6G antibodies) or IFN γ antibody-depleted (i.p. injection of anti-IFN γ antibodies) adult WT mice. **L**, CXCL1 and TNF- α concentrations after ex vivo LPS stimulation. Data were analysed using one-way ANOVA followed by Dunn's multiple comparisons test. * $p < 0.05$, *** $p < 0.001$.

determining the immunological outcome. While we show here that β -glucan aggravates LPS-induced immunopathology, repeated challenges of LPS has also long-been associated with a dampening of immune responses, termed tolerance^{31,32}. However, ambient exposure of LPS has also been linked with training leading to enhanced bacterial clearance²⁵. Evidently, the impacts of trained immunity are intricate and context dependent. We postulate two signals are required for the induction of trained immunity: signal 1, which is essential for reprogramming innate immune cells centrally (HSCs) or peripherally (e.g. AMs), and signal 2, an environmental cue critical for activating their functional capacity. In this study, we provide evidence supporting this conceptual framework.

β -glucan-trained AMs were able to generate a robust response both *in vitro* and *in vivo* to bacterial (LPS) or viral (poly(I:C)) ligands. The increased response in β -glucan-trained AM was maintained up to 28 days, which is an indication of long-term reprogramming of AMs. The reprogramming of trained AMs was supported by significant alterations in both the transcriptomic and metabolic states. Using adoptive transfer experiments, we then showed that the functional reprogramming of β -glucan trained AMs persisted even in the absence of the initial systemic β -glucan administration. This finding highlights the persistence of the intrinsic training program in AMs.

AMs are originated from yolk sac and fetal liver monocytes, which seeds the lung as soon as we take our first breath of air¹⁰. Importantly, AMs are the first immune cell to respond to any particles or pathogens that reach the lower airways of the lungs, thus alterations in their functional capacity can significantly impact subsequent immune responses. For instance, given AMs are in direct contact with surfactant (produced by alveolar type 2 epithelial cells) and invading pathogens, a defect in GM-CSF or TGF β signalling leads to accumulation of surfactant (alveolar proteinosis) and increased susceptibility to pulmonary infections^{33,34}. AMs are able to sense and integrate multiple environmental signals - such as pH³⁴, temperature^{35,36}, osmolarity³⁷, metabolites including fatty acids^{38,39}, extracellular membrane components and danger signals⁴⁰ - to maintain tissue homeostasis⁴¹. However, the mechanisms underlying their ability to adapt to environmental stimuli while maintaining lung tissue homeostasis without impairing gas exchange is largely unknown⁴². AMs exhibit a remarkable plasticity, demonstrating a spectrum of functional polarization that ranges from regulatory to pro-inflammatory and anti-inflammatory states. For instance, AM anti-inflammatory polarization in a LPS model of sepsis was TNF α dependent, as AMs exposed to TNF α exhibited diminished phagocytosis, superoxide anion (O₂⁻) and CXCL1 production, with reduced neutrophils recruitment³². Consequently, these mice had a reduction in lung clearance of *P. aeruginosa* infection. Influenza virus was able to induce similar anti-inflammatory function in AMs with decreased CXCL1 production and neutrophils recruitment via type I IFN pathway⁴³, leading to an increased susceptibility to super bacterial infection. On the other hand, AMs appear to be more resistant to polarization towards a pro-inflammatory state⁴⁴. A recent study demonstrated that exposure of lungs to ambient amount of LPS trains AMs in type I IFN dependent, but type II IFN and T cell independent manner²⁵. However, in our model system, systemic administration of β -glucan trains AMs in a type II IFN dependent, but type I IFN independent manner. Similarly, in live infection models, we and others have identified IFN γ signaling a key player in AM training after BCG vaccination²⁶, influenza infection⁴⁵, pneumococcal infection⁴⁶ and intranasal infection with an adenoviral vector⁸. Although we have not identified which cells produce IFN γ in the β -glucan model, it has been demonstrated by our group and others that following BCG vaccination, CD4⁺ T cells are the major source of IFN γ ²⁶, whereas after pulmonary adenovirus infection, CD8⁺ T cells predominantly produce IFN γ ²⁷. Influenza infection has also been described to induce IFN γ -dependent AM training with NK cells being the major source⁴⁵. Interestingly, we found that neutrophils were also required for β -glucan-mediated AMs training. We have been recently shown that β -glucan can reprogram HSCs via Dectin 1 and type I IFN signaling to promote granulopoiesis and the generation of trained neutrophils promoting disease tolerance to influenza infection²⁰. Although we have shown that the recruitment of these trained neutrophils and type II IFN signaling were required for training AMs, the cellular and

molecular mechanisms involved in this dialogue is unknown and requires further investigation. Interestingly, we have recently demonstrated that, in addition to GM-CSF and TGF β signalling, neutrophils are critical for AMs self-renewal and maintenance during early lung development via the production of 12-hydroxyeicosatetraenoic acid (12-HETE)⁴⁷. The absence of 12-HETE leads to a significant reduction in the number AMs in adult lungs, enhanced senescence, and consequently increased susceptibility to IAV or SARS-CoV-2 infection. Thus, there might be a constant bidirectional dialogue between neutrophils and AMs, with neutrophils providing cues from internal organelles to AMs, and AMs offering signals from the external environment to neutrophils, which then return to their graveyard in the BM.

Although we have not directly tested the contribution of circulating monocytes in the initial reprogramming of AMs via β -glucan, the persistence of trained immunity by adoptively transferred AMs into *Csf2rb*^{-/-} mice suggests that the maintenance of the trained AM state was independent of bone marrow derived monocytes. Additionally, the findings from Theobald *et al* indicate that Dectin-1 is the receptor responsible for recognizing β -glucan which has been shown to activate macrophages and induce trained immunity in several models^{20, 48, 49, 50}. Here we found that β -glucan-mediated AM training, when administered systemically, was independent of Dectin-1. There are two potential explanations for this observation. First, it has been shown that β -glucan can also activate signaling via other TLRs, particularly TLR2 or complement receptor 3 (CR3)^{51, 52, 53}, and second as the β -glucan is a particulate, its internalization by phagocytes can also initiate signaling^{54, 55, 56}. Additionally, our understanding of how administration of β -glucan in peritoneal cavity leads to HSCs training in the BM and AMs training in thoracic cavity is extremely limited. Thus, addressing the molecular mechanisms of β -glucan signaling pathways (e.g., Dectin1-dependent and independent) in both immune and non-immune cells, as well as its mode of action (e.g., direct access to an organ versus indirect effects via systemic release of cytokines), is necessary to delineate the deleterious versus protective effects of β -glucan-mediated trained immunity.

β -glucan is present in the cell wall of all fungi but will vary between different species and strains. In fact, the differential 1,3 to 1,6 glycosidic branching and molecular weight significantly impacts the response to the compound with a large variability of scientific findings contingent on the type of β -glucan used in a study^{57, 58, 59}. Fungi make up a portion of the human microbiome, termed the mycobiome. Several studies have described a high mycotic diversity between different populations, and even significant variability within an individual overtime^{60, 61, 62}. The variance in gut colonization by fungal species can moreover cause gut dysbiosis which has been associated with poorer outcomes during SARS-CoV-2 infection, sepsis and cancer immunotherapy^{63, 64, 65, 66, 67}. It is tempting to postulate that the levels of the mycobiome as well as its composition can influence subsequent immune reactions partly due to a distinct β -glucan makeup. In fact, antibodies to various types of β -glucan was detected in adult sera with different levels correlating with a person's occupation⁶⁸. Thus, the quantity and quality of circulating β -glucan in an individual at steady state can remarkably affect the subsequent immune responses to sterile or microbial inflammation. The concept of detrimental training by endogenous agents has also been described with heme, which similarly aggravated LPS-induced inflammation⁶⁹. Basal levels of such immunomodulatory molecules provide a basis for host response heterogeneity and corresponding excessive inflammation⁷⁰. Understanding these underlying processes may provide important insights for developing novel therapeutic approaches against immune-mediated pulmonary disease.

Materials and methods

Mice

C57BL/6, *Csf2rb*^{-/-}, *Dectin1*^{-/-}, *IfnaR*^{-/-} and *IfngR*^{-/-} mice were purchased from Jackson Laboratories. All animals were housed and inbred at the animal facility of the Research Institute of McGill University under specific pathogen-free conditions with *ad libitum* access to food and water, a temperature of 21 °C (±1 °C), relative humidity of 40–60% (±5%) and light cycle of 12 h on, 12 h off (daily cycle). *Sex as a biological variable*: Mice were randomly allocated to experimental groups, and experiments were performed using both female and male age- and sex-matched mice. Similar findings were reported for both sexes.

β-glucan training

Mice were administered intraperitoneally with 1mg of β-1,3-glucan purified from *Saccharomyces cerevisiae* (Sigma) diluted in 100μL of PBS seven days or twenty-eight days before lung injury or AMs collection for *ex vivo* stimulation.

Acute lung injury models

Mice were administered with 50μg of *Escherichia coli* O55:B55 LPS (Sigma) or poly(I:C) (Invivogen) in PBS (25 μL per mouse, intranasally) to induce TLR-4 or 3-mediated acute lung injury (ALI), respectively.

Lung microCT scan

The trachea was cannulated with a 22-gauge cannula and an intra-thoracic pressure of 25mmH₂O was generated using a manometer. Images were acquired right after lung inflation using the nanoScan SPECT + CT (Mediso®) allowing a resolution of 20μm. DICOM software was used to analyse the microCT scans determining the average lung Hounsfield unit (HU) and the proportion of non- or poorly-aerated lung (HU -500; +100).

BAL and lung collection

Broncho-alveolar lavage (BAL) samples collected by cannulating the trachea with a 22-gauge cannula, then washing the lungs with 3× 1 mL of cold, sterile PBS. The total volume of the recovered fluid after lavage was around 0.7 ml. Samples were centrifuged (1,500 r.p.m., 10 min). Lung tissues were perfused with 10 mL of PBS, collected and minced before collagenase type I (3mg, Worthington CLS-1), elastase (3mg, Worthington ESL) and DNase I (0.4mg, Worthington D) digestion for 30min at 37 °C. Lungs were filtered through a 70 μm nylon mesh, and red blood cells were lysed.

Endothelial permeability

LPS or poly(I:C)-challenged mice were intravenously injected with 400 μl of Evan's blue dye (2% in PBS) into the mice. After 1 h, mice were euthanized, and lungs were perfused with 10 ml of PBS. Evan's blue then extracted by overnight incubation in formamide at 56 °C (lungs) and quantified by spectrophotometry analysis using a standard curve of Evan's blue in formamide.

Histopathological analysis

Histopathological analysis was performed as previously described⁴⁷. Lungs were inflated and fixed for 48 h with 10% formalin, and then embedded in paraffin. Sections (5 μm) were cut and stained with haematoxylin and eosin. Slides were scanned at a resolution of ×40 magnification, and pictures were taken using a Leica Aperio slide scanner (Leica). Histology samples were evaluated according to ATS 2011 guidelines regarding “Features and measurements of experimental acute lung injury in animals” by a blinded observer.

ELISA

CXCL1, TNF- α , IL-6 and IFN γ levels in BAL were assessed by ELISA (R&D Systems).

Protein in BAL

Samples were centrifuged (1,500 r.p.m., 10 min), and total protein content was assessed using a Pierce BCA Protein assay (ThermoFisher).

Flow cytometry

BAL and total lung cell counts were determined with a haemocytometer, and 1–2 million cells were used for staining. Cells were initially stained with viability dye e506 (Invitrogen, 20 min, 4 °C) and surface stained with anti-CD16/32 (BD Bioscience) in 0.5% BSA/PBS solution to block nonspecific AB interaction with Fc receptors (10 min, 4 °C). Cells were then surface-stained with different combinations of PE-Cy7-conjugated anti-CD11c, BUV786-conjugated anti-Siglec-F, BUV395-conjugated anti-CD11b, APC-Cy7-conjugated anti-Ly6G, BUV737-conjugated anti-CD45.2 antibodies (all from BD Biosciences). For IFN γ intra-cellular staining, cells were fixed and permeabilized using BD CytoFix/CytoPerm (BD Bioscience) before intracellular staining with PE-conjugated anti-IFN γ antibodies (BD Biosciences). Flow cytometry was performed using a BD LSR Fortessa X-20 instrument (BD Biosciences) with FACSDiva software v.8.0.1 (BD Biosciences). Analysis was performed using FlowJo software v.10.7.1 (Tree Star).

Intravascular staining

In vivo discrimination between pulmonary vasculature and parenchyma was performed as previously described⁴⁷. Adult WT mice were given 2 μ g of FITC-conjugated anti-CD45.2 intravenously to label all circulating cells. Three minutes later, mice were euthanized and lungs collected, stained *ex vivo* with BUV395-conjugated anti-CD45.2 antibody to determine the parenchymal (cells only labelled with the *ex vivo* antibody) or vascular localization of the cells (cells labelled with both antibodies).

AM depletion

WT mice were treated with control or clodronate liposomes (70 μ L, intranasally; Liposoma BV). The LPS-induced ALI was then performed at day 2 after clodronate instillation.

Adoptive transfer models

AMs from WT mice which received i.p. PBS or β -glucan were collected as described above and resuspended at a density of 5×10^4 cells per 5–7 μ L of RPMI1640 medium supplemented with 10% (v/v) FBS, 2 mM l-glutamine, 10 mM HEPES and 100 U ml⁻¹ penicillin–streptomycin. AMs were then transferred by the intranasal route into Day 2 *Csf2rb*^{-/-} pups. LPS-induced ALI was performed 6 weeks after AM adoptive transfer. BAL and lung tissue were collected and processed as described above for endothelial permeability, flow cytometry, total BAL cytokine and protein content experiments.

Isolation and culture of alveolar macrophages

AMs were collected by BAL of naive mice using cold, sterile PBS (5 $\times$ 1 mL for adult mice). AMs were cultured in the specific media described above. After 1 h of adhesion, AMs were washed with PBS and placed in fresh medium.

Ex vivo stimulation

AMs from WT, *Dectin1*^{-/-}, *IfnaR*^{-/-} and *IfngR*^{-/-} mice which received i.p. PBS or β -glucan were collected as described above and 5×10^4 cells in specific media were distributed per well. AMs were stimulated with 50 ng/ mL of LPS (Sigma) for 4 hours at 37°C.

Library preparation and RNA-seq

Total RNA was collected from BAL AMs from four WT mice per conditions (i.p. PBS, no *ex vivo* stimulation / i.p. β -glucan, no *ex vivo* stimulation / i.p. PBS, *ex vivo* LPS stimulation / i.p. β -glucan, *ex vivo* LPS stimulation). After RNA quality controls, sequencing libraries were constructed using the Illumina TruSeq protocol. Libraries were sequenced on an Illumina NovaSeq 6000 (paired-end 100 base pair) to an average depth of 51,189,336 reads per sample.

RNA-seq data analysis

RNA-seq reads were aligned to the *Mus musculus* genome from Ensembl version 99 using STAR (version 2.7.3a) was used⁷¹. All regions overlapped between referenced exons and alignments were counted using featureCounts (subread-1.6.4)⁷². Low abundance genes were filtered out leaving 12,894 genes for subsequent analysis.

Differential expression analyses were performed using the DESeq2 package (DESeq2 1.40.2)⁷³. Gene ontology analysis was realized with the R package TopGO. For gene set enrichment analysis (GSEA) a ranked list of the differentially expressed genes was used with clusterProfiler v4.8.2⁷⁴, and the Molecular Signatures Database MSigDB v7.5.1.

Extracellular flux analysis

Seahorse assay of isolated cells was performed as previously described⁴⁷. Real-time OCRs of AMs were measured in XF medium (non-buffered RPMI containing 2 mM l-glutamine, 25 mM glucose and 1 mM sodium pyruvate) using a Seahorse Xfe 96 Analyzer (Agilent Technologies). For the mitochondrial stress test, mitochondrial inhibitors oligomycin (1.5 μ M), fluorocarbonyl cyanide phenylhydrazone (FCCP) (1 μ M), antimycin A and rotenone (0.5 μ M) were used as per the manufacturer's recommendations. In brief, cells were seeded at a density of 100,000 cells per well and 3 basal measurements were taken. Following this, two consecutive measurements were taken following each injection of oligomycin, FCCP and antimycin A with rotenone. All measurements were normalized to cell number using crystal violet dye extraction assay. Oxygen consumption curves, OCRs and ECARs were generated using Wave Desktop 2.3 (Agilent Technologies).

IFN γ and neutrophils depletion experiments

Sex- and age-matched adult WT mice received intra-peritoneal 200 μ g of anti-IFN γ (rat IgG1k, Biolegend), anti-Ly6G (rat IgG2a,k, Biolegend) or appropriated control isotypes injection at Day-1, Day 0, Day 2, Day 4 and Day 6 per intraperitoneal injection of β -glucan. AMs collection for *ex vivo* stimulation was performed at Day 7 after intraperitoneal β -glucan injection.

Ethics statement

All experiments involving animals were approved by the McGill University Animal Care Committee (permit number 2010–5860) in accordance with the guidelines set out by the Canadian Council on Animal Care.

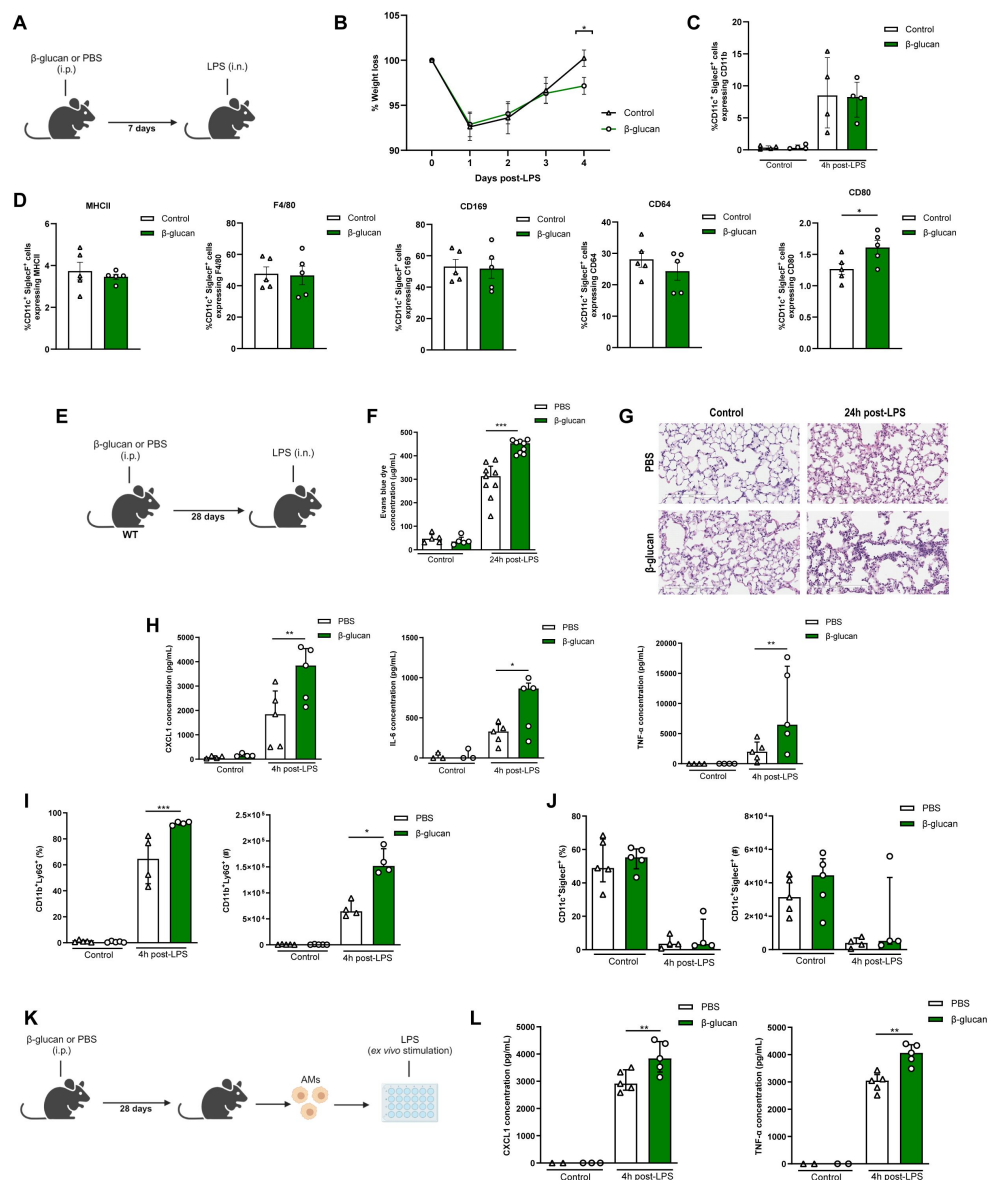
Statistical analysis

Data are presented as the median with interquartile range. Statistical analyses were performed using GraphPad Prism v.9.1.2 software (GraphPad). Statistical differences were determined using one-way analysis of variance (ANOVA) followed by Dunn's multiple comparisons test, paired or unpaired two-tailed t-test or two-tailed Mann–Whitney test. Differential gene expression analysis was carried out using DEseq2 package.⁷³ [DOI](#)

Data availability

All data supporting the findings of this study are included in the published article and supplementary materials. Bulk RNA-seq data have been deposited to the European Nucleotide Archive and will be made publicly available upon publication. Source data are provided with this paper.

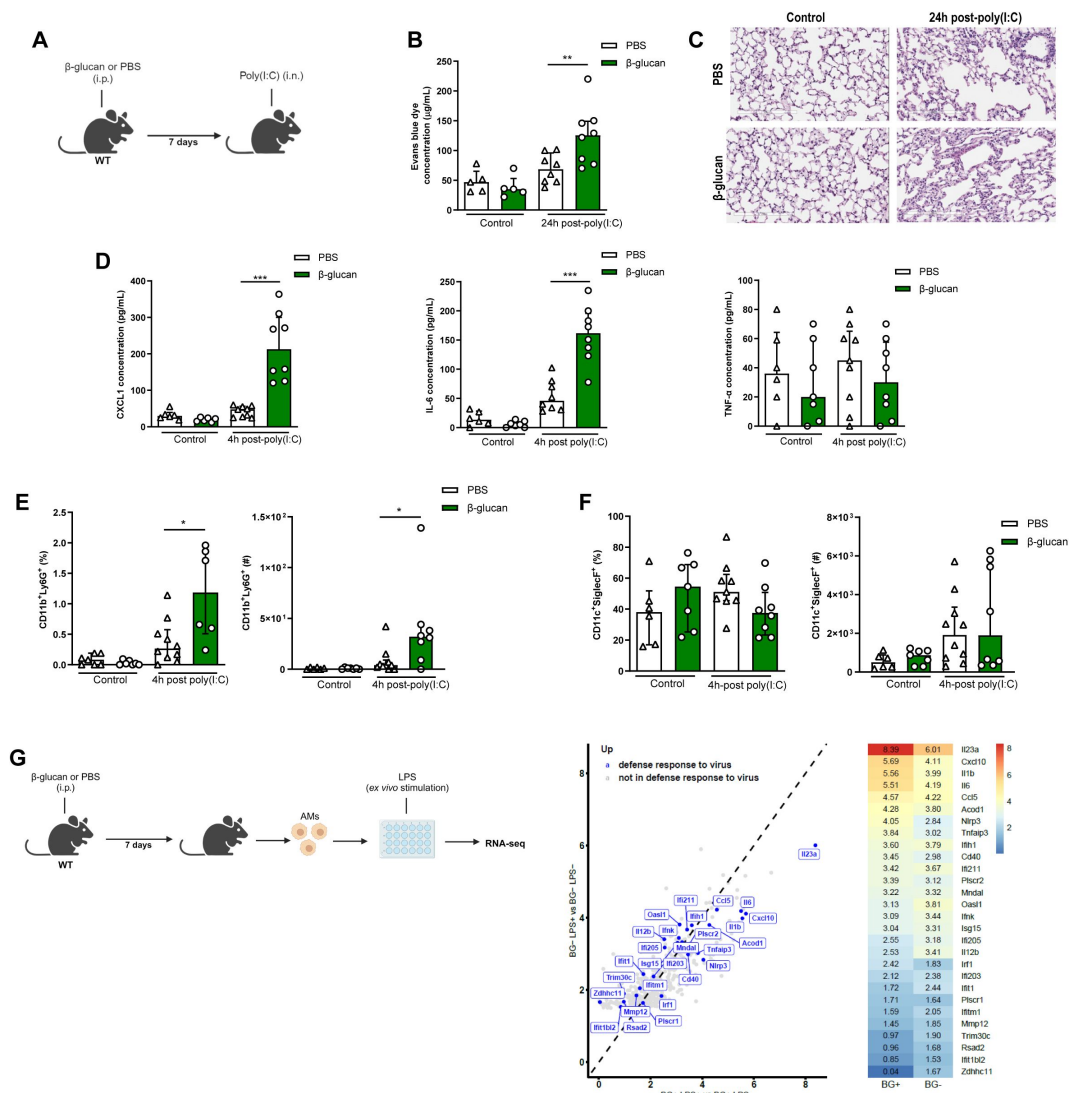
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Supplementary Figure 1.

Long-term effects of β-glucan-mediated trained immunity on LPS-induced ALI.

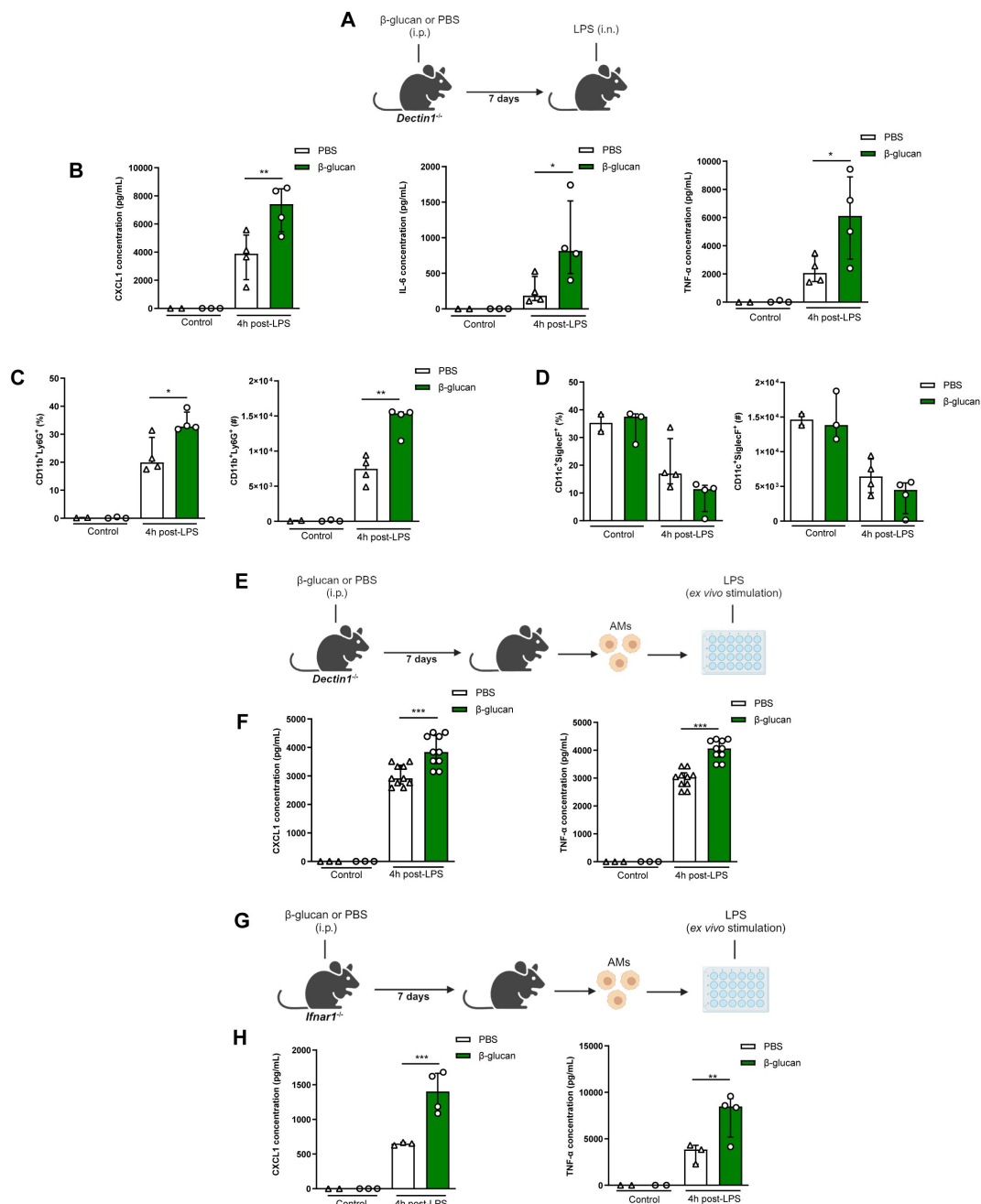
A, Schematic of the β-glucan-induced training seven days before lipopolysaccharide (LPS)-induced acute lung injury (ALI) model for weight loss monitoring. **B**, Weight loss curve post-LPS instillation (n=10). **C**, CD11b expression on CD11c⁺SiglecF⁺ cells in the bronchoalveolar lavage (BAL) (n=4). **D**, Expression of AM-associated marker (left to right: MHCII, F4/80, CD169, CD64, CD80) on CD11c⁺SiglecF⁺ cells in the bronchoalveolar lavage (BAL) (n=5). **E**, Schematic of the β-glucan-induced training twenty-eight days before lipopolysaccharide (LPS)-induced acute lung injury (ALI) model. Experiments were performed in sex- and age-matched 10-12 weeks old control (i.p. PBS, white bars) and trained (i.p. β-glucan, green bars) WT mice. **F**, Alveolar capillary membrane permeability assessed by lung Evans blue dye concentration. **G**, Lung histology after staining with haematoxylin and eosin. **H**, BAL chemokine C-X-C motif ligand 1 (CXCL1) concentration (left) and pro-inflammatory cytokines (IL-6: interleukin-6 (middle) and TNF-α: tumor necrosis factor α (right)). **I**, Quantification of BAL neutrophils frequency (left) and absolute count (right) (gated on single live cells, CD45.2⁺, CD11c⁺, Siglec-F⁺, CD11b⁺, Ly6G⁺). **J**, quantification of BAL alveolar macrophages frequency (left) and absolute count (right) (gated on single live cells, CD45.2⁺, CD11c⁺, Siglec-F⁺). **K**, Schematic of control (i.p. PBS, white bars) or 28 days β-glucan-trained (i.p. β-glucan, green bars) AMs collected from adult WT mice *ex vivo* stimulation with LPS. **L**, Chemokine C-X-C motif ligand 1 (CXCL1) and tumor necrosis factor α (TNF-α) concentrations after *ex vivo* LPS stimulation. Data were analysed using unpaired t-test and one-way ANOVA followed by Dunn's multiple comparisons test. * p<0.05, ** p<0.01, *** p<0.001.



Supplementary Figure 2.

β -glucan-mediated trained immunity increases poly(I:C)-induced ALI.

A, Schematic of the β -glucan-induced training seven days before poly(I:C)-induced acute lung injury (ALI) model. Experiments were performed in sex- and age-matched 10-12 weeks old control (i.p. PBS, white bars) and trained (i.p. β -glucan, green bars) WT mice. **B**, Alveolar capillary membrane permeability assessed by lung Evans blue dye concentration. **C**, Lung histology after staining with haematoxylin and eosin. **D**, BAL chemokine C-X-C motif ligand 1 (CXCL1) concentration (left) and pro-inflammatory cytokines (IL-6: interleukin-6 (middle) and TNF- α : tumor necrosis factor α (right)). **E**, Quantification of BAL neutrophils frequency (left) and absolute count (right) (gated on single live cells, CD45.2⁺, CD11c⁻, Siglec-F⁻, CD11b⁺, Ly6G⁺). **F**, Quantification of BAL alveolar macrophages frequency (left) and absolute count (right) (gated on single live cells, CD45.2⁺, CD11c⁺, Siglec-F⁺). **G**, Schematic of control (i.p. PBS, white bars) β -glucan-trained (i.p. β -glucan, green bars) AMs collected from adult WT mice *ex vivo* stimulation with LPS. Differential expression of viral defense genes in response to LPS in β -glucan-trained AMs. Data were analysed using one-way ANOVA followed by Dunn's multiple comparisons test. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.



Supplementary Figure 3.

β -glucan-mediated AM reprogramming is independent of Dectin-1 and type I interferon signaling.

A, Schematic of the β -glucan-induced training seven days before lipopolysaccharide (LPS)-induced acute lung injury (ALI) model. Experiments were performed in sex- and age-matched 10-12 weeks old control (i.p. PBS, white bars) and trained (i.p. β -glucan, green bars) *Dectin1*^{-/-} mice. **B**, BAL chemokine C-X-C motif ligand 1 (CXCL1) concentration (left) and pro-inflammatory cytokines (IL-6: interleukin-6 (middle) and TNF- α : tumor necrosis factor α (right)). **C**, Quantification of BAL neutrophils frequency (left) and absolute count (right) (gated on single live cells, CD45.2⁺, CD11c⁻, Siglec-F⁻, CD11b⁺, Ly6G⁺). **D**, Quantification of BAL alveolar macrophages frequency (left) and absolute count (right) (gated on single live cells, CD45.2⁺, CD11c⁺, Siglec-F⁺). **E**, Schematic of control (i.p. PBS, white bars) β -glucan-trained (i.p. β -glucan, green bars) AMs collected from adult *Dectin1*^{-/-} mice *ex vivo* stimulation with LPS. **F**, Chemokine C-X-C motif ligand 1 (CXCL1) and tumor necrosis factor α (TNF- α) concentrations after *ex vivo* LPS stimulation. **G**, Schematic of control (i.p. PBS, white bars) β -glucan-trained (i.p. β -glucan, green bars) AMs collected from adult *Ifnar1*^{-/-} mice *ex vivo* stimulation with LPS. **H**, Chemokine C-X-C motif ligand 1 (CXCL1) and tumor necrosis factor α (TNF- α) concentrations after *ex vivo* LPS stimulation. Data were analysed using one-way ANOVA followed by Dunn's multiple comparisons test. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

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Additional information

Author contributions

Conceptualization: R.P., E.P. and M.D. Methodology: R.P., E.P., A.S.K., M.S., K.A.T., R.S., E.L., L.J., J.P. and M.D. Investigation: R.P., K.A.S., E.P., J.P. and M.D. Funding acquisition: M.D. Project administration: R.P. and M.D. Supervision: E.P., J.P. and M.D. Writing original draft: R.P. K.A.T and M.D. Writing, review, and editing: K.A.T, R.P., E.P., K.A.S., J.P. and M.D.

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Reviewer #1 (Public review):

Summary:

The concept that trained immunity, as defined, can be beneficial to subsequent immune challenges is important in the broad context of health and disease. The significance of this manuscript is the finding that trained immunity is actually a two-edged sword, herein, detrimental in the context of LPS-induced Acute Lung Injury that is mediated by AMs.

Strengths:

Several lines of evidence in different mouse models support this conclusion. The postulation that differences in immune responses in individuals is linked to differences in the mycobiome and consequent B-glucan makeup is provocative.

Weaknesses:

However, the findings that the authors state are relevant to sepsis are actually confined to a specific lung injury model and not classically-defined sepsis, the ontogeny of the reprogrammed AMs is uncertain, and links in the proposed signaling pathways need to be strengthened.

Comments on the latest version:

The manuscript is improved with further clarifications and additional experimentation. My prior concerns are addressed.

<https://doi.org/10.7554/eLife.102068.2.sa2>

Reviewer #2 (Public review):

Summary:

Prével et al. present an in vivo study in which they reveal an interesting aspect of β -glucan, a known inducer of enhanced immune responses termed trained immunity in sterile inflammation. The authors can show that β -glucan's can reprogram alveolar macrophages (AMs) in the lungs through neutrophils and IFN γ signaling and independent of Dectin1. This reprogramming occurs at both transcriptional and metabolic levels. After β -glucan training, LPS induced sterile inflammation exacerbated acute lung injury via enhanced immunopathology. These findings highlight a new aspect of β -glucan's role in trained immunity and its potential detrimental effects when enhanced pathogen clearance is not required.

Strengths:

- This manuscript is well-written and effectively conveys its message.
- The authors provide important evidence that β -glucan training is not solely beneficial but depending on the context can also enhance immunopathology. This will be important to the field for two reasons. It shows again that trained immunity can also be harmful. Jentho et al. 2021 had already provided further evidence for this aspect. And it highlights anew that LPS application is an insufficient infection model.

Original weaknesses noted:

- Only a little physiological data from the in vivo models is provided.
- Effects in histology appear to be rather weak.

Comments on latest version:

The authors have revised the new version according to my suggestions or responded in a sufficient manner to my requests, with one exception. I recommend to rename TNF as explained by Grimstad in JAMA Dermatol. 2016;152(5):557.

<https://doi.org/10.7554/eLife.102068.2.sa1>

Author response:

The following is the authors' response to the original reviews

Reviewer #1 (Public review):*Summary:*

The concept that trained immunity, as defined, can be beneficial to subsequent immune challenges is important in the broad context of health and disease. The significance of this manuscript is the finding that trained immunity is actually a two-edged sword, herein, detrimental in the context of LPS-induced Acute Lung Injury that is mediated by AMs.

Strengths:

Several lines of evidence in different mouse models support this conclusion. The postulation that differences in immune responses in individuals are linked to differences in the mycobiome and consequent B-glucan makeup is provocative.

Weaknesses:

The findings that the authors state are relevant to sepsis, are actually confined to a specific lung injury model and not classically-defined sepsis. In addition, the ontogeny of the reprogrammed AMs is uncertain. Links in the proposed signaling pathways need to be strengthened.

Reviewer #2 (Public review):*Summary:*

Prével et al. present an in vivo study in which they reveal an interesting aspect of β -glucan, a known inducer of enhanced immune responses termed trained immunity in sterile inflammation. The authors can show, that β -glucan's can reprogram alveolar macrophages (AMs) in the lungs through neutrophils and IFN γ signaling and independent of Dectin1. This reprogramming occurs at both transcriptional and metabolic levels. After β -glucan training, LPS-induced sterile inflammation exacerbated acute lung injury via enhanced immunopathology. These findings highlight a new aspect of β -glucan's role in trained immunity and its potential detrimental effects when enhanced pathogen clearance is not required.

Strengths:

(1) This manuscript is well-written and effectively conveys its message.

(2) The authors provide important evidence that β -glucan training is not solely beneficial, but depending on the context can also enhance immunopathology. This will be important to the field for two reasons. It shows again, that trained immunity can also be harmful. Jentho et al. 2021 have already provided further evidence for this aspect. And it highlights anew that LPS application is an insufficient infection model.

Weaknesses:

(1) Only a little physiological data is provided by the in vivo models.

(2) The effects in histology appear to be rather weak.

Reviewer #1 (Recommendations for the authors):

The opening paragraph in the introduction focuses on sepsis. This is misleading since this manuscript does not address sepsis but rather intranasal-administered LPS-induced acute lung injury.

We are in total agreement with the reviewer and have modified the introduction to focus on acute lung injury with clinical relevance more associated to TLR4-mediated acute lung injury and lung inflammation.

The authors make definitive statements that AMs originate from fetal liver monocytes. However, it is well known that the ontogeny of AMs is complex and AMs can be populated, in part, from peripheral monocytes. The ontogeny of reprogrammed AMs was not addressed in this study but they may come from monocyte-derived AMs following β -glucan training (transfer of AMs into Csf2rb KO mice does not prove the contrary). In this regard, do, for example, the percentages of CD11b⁺ AMs change? More phenotyping of the control and reprogrammed AMs would enhance the interpretation of the findings.

The reviewer is correct that the ontogeny of AMs can be heterogenous, especially following a pulmonary challenge. In β -glucan-treated mice, Figure 1I shows no changes in frequency or number of AMs in the BAL. As the reviewer suggested, we repeated this experiment and incorporate more markers for AMs. New Supplementary Figure 1C shows the expression of CD11b on AMs (CD11c⁺SiglecF⁺) from control and β -glucan-treated mice. While the frequency increases with LPS administration, we show no difference between control and β -glucan groups suggesting β -glucan does not induce the expansion of monocyte-derived AMs. Additionally, in New Supplementary Figure 1D, we show the expression of AM-associated markers in order to better delineate their phenotype. We observed no differences in MHCII, CD169, CD64 and F4/80 in β -glucan-treated mice, but an increase in CD80⁺ AMs following β -glucan suggesting enhanced activation corroborating their proinflammatory phenotype. Collectively, these data indicate that while the frequency and number of either yolk-sac or BM-derived AMs are unchanged in the β -glucan treated mice, the activation of AMs is enhanced after the systemic treatment with β -glucan.

The abstract seems to overpromise a bit. First, it mentions trained immunity and HSCs, but they don't seem to formally address either in the context of this model (there is reprogramming as assessed by transcriptome and metabolic analyses which is suggestive as stated by the authors, but do the changes overlap significantly with classically trained immunity?), and second, it links phenotypes together in a pathway(s) that they haven't actually interrogated - although they look at transcripts and do a Seahorse assay they don't actually confirm that any of those findings are related to the increased response to LPS in vivo. The long discussion with all the caveats highlights these limitations, all relegated to future studies.

We thank the reviewer for this comment. In response, we have revised the abstract to more accurately highlight the key findings of this study. Specifically, we introduced the concept of central trained immunity to describe the phenomena commonly observed with β -glucan treatment, contrasting it with the peripheral trained immunity detailed in the manuscript.

The use of Csf2rb^{-/-} mice to complement the clodronate approach is interesting (this approach has been used in the past with influenza virus). In addition to lacking AMs, these mice develop pulmonary alveolar proteinosis. Do the authors have histopathology from these mice in the current model? They mention PAP in the discussion.

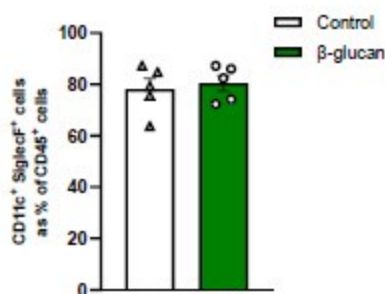
Pulmonary alveolar proteinosis (PAP) typically develops in *Csf2b*^{-/-} mice from 12 weeks of age onwards (Stanley et al., Proc Natl Acad Sci USA, 1994). However, in our model, mice were euthanized at 6 weeks, ensuring that pulmonary function and structure remained intact. A hallmark of PAP is the accumulation of protein, primarily surfactant, in BAL. To investigate this, we measured BAL protein concentration and observed no differences at baseline (Figure 2F). These findings were further supported by the absence of differences in BAL proinflammatory cytokine concentrations (Figure 2H).

A question about their BAL technique? In the control mice without glucan/LPS stimulation, only 40% of BAL cells are AMs [and the total number of AMs (range of <103 to 2-3 x 104) is at least 5-fold lower than typically seen in BALs from healthy mice (105), and there didn't seem to be many PMNs either. Are 60% of the BAL cells lymphocytes/RBCs? Is it possible that overall AM numbers are changing, but CD11c/SiglecF-positive cell numbers stay the same (only assessed 2 markers)? More phenotyping would help.

We appreciate the reviewer's comment and would like to clarify that alveolar macrophages (AMs) are presented in the manuscript as a frequency of viable cells rather than as a frequency of CD45⁺ cells, to ensure consistency throughout the study. The remaining cells in the samples are likely epithelial cells and lymphocytes, as red blood cells are lysed during sample processing. For additional context, we now provide data showing AMs as a percentage of CD45⁺ cells, which account for 80–90% of leukocytes. Furthermore, in New Supplementary Figure 1D, we highlight the expression of AM-associated markers to better define their phenotype. We observed no differences in MHCII, CD169, CD64, or F4/80 expression in β -glucan-treated mice. However, there was an increase in CD80⁺ AMs, indicating enhanced activation and corroborating their proinflammatory phenotype.

Author response image 1.

AMs as percentage of CD45⁺ cells. Mice were treated with β -glucan for seven days. We show CD11c⁺SiglecF⁺ cells in the bronchoalveolar lavage (BAL) as a percentage of CD45⁺ cells (n=5).



Line 130-131. TNF is decreased and not pointed out.

In the poly(I:C) model, the difference in the BAL TNF concentration is not statistically different between naïve and trained mice due to high variability of data. The reviewer is correct that TNF α does not appear to reflect Poly(I:C)-mediated ALI. We have included this point in the revised manuscript (Line 146-148).

Reviewer #2 (Recommendations for the authors):

Suggestions:

(1) The authors provide evidence for enhanced ALI via different techniques, e.g. histology, vascular leakage, immune cell composition in BAL etc. It would be interesting to see whether there were any changes in the disease severity of ALI. If possible the authors could provide data for survival, temperature, weight, and/or glucose in the different groups.

Mice are extremely resistant to the pulmonary LPS model. We have previously assessed lethality of our LPS model, and all mice survive even with an increased intranasal dose of LPS 200µg (Pernet et al, Nature, 2023). To address the reviewer concerns, we next assessed the morbidity by monitoring weight loss following LPS challenge and showed β-glucan-treated mice exhibit a delayed recovery time after 4 days LPS treatment (New Supplementary Figure 1B).

(2) The authors show that β-glucan mediated training enhances ALI. Conversely, the opposite, decreased immunopathology should be observed in case an LPS tolerance model would be used. I am wondering whether this has already been performed, given that the (LPS/immune)tolerance field is already older than the training field. If not, I suggest incorporating this feature in their discussion.

Thank you for this insightful comment. While LPS has long been recognized to induce tolerance, studies have also shown that intranasal exposure to ambient levels of LPS can induce alveolar macrophage (AM) training via type I interferon signaling (Zahalka et al., Mucosal Immunol, 2022). In contrast, Mason et al. demonstrated that systemic LPS stimulation induces tolerance through TNF-α signaling, resulting in diminished AM phagocytosis and superoxide production. This leads to reduced neutrophil recruitment and impaired bacterial clearance in a *Pseudomonas aeruginosa* pneumonia model (J Infect Dis, 1997). Furthermore, we recently reported that systemic administration of β-glucan induces central trained immunity, generating a distinct subset of regulatory neutrophils that promote disease tolerance against influenza viral infection (Khan et al., Nat Immunol, 2025). These findings highlight the complex and context-dependent interplay between training and tolerance. We have expanded on this point in the discussion section of the revised manuscript (Lines 289-297).

(3) The finding that trained immunity can exert not only beneficial effects but also enhance immunopathology is interesting and should be further explored. Already Jentho et al. (PNAS 2021) have shown that upon sterile inflammation as imposed by LPS, (heme) training can lead to enhanced mortality. This might be a relevant trade-off in trained immunity since no beneficial resistance effect by pathogen killing can be obtained. It would be interesting to see, in their model, whether heme would also enhance ALI after intranasal LPS application. Or at least, can the authors discuss this finding more, also in relation to the already published evidence?

Thank you for raising this interesting point, which is indeed relevant to our study. Jentho et al. demonstrated that training by heme can be beneficial in combating infectious challenges but can have deleterious effects in the context of sterile inflammation. The concept of endogenous training agents like heme, with their diverse effects on immune cells, aligns well with our βglucan model, particularly given the high prevalence of fungal agents in the microbiome.

While investigating the effects of heme on alveolar macrophages would certainly be intriguing, Jentho and colleagues have already reported the maladaptive effects of heme, such as tissue damage, during sterile LPS-induced inflammation. As such, these findings might be redundant in the context of our model. However, we have drawn a relevant parallel and expanded on this discussion in the revised manuscript (Lines 382-385).

(4) It is not clear how the histologies were evaluated. This is a field of great subjectivity. The authors should describe it in more detail. The best option would have been a blinded observer. Was this done?

Histology samples were evaluated according to ATS 2011 guidelines regarding “Features and measurements of experimental acute lung injury in animals” by a blinded pathologist. We have specified this in the methods of the revised manuscript.

Minor:

(1) Line 108 and ff. Please change TNF, not TNFa

Since we used an ELISA specific for TNF- α rather than general TNF, it is more accurate to refer to it as TNF- α .

(2) Line 513 and ff. Please use Greek letters when appropriate, e.g. IFN- γ not IFNg.

Thank you for pointing out these mistakes, we rectified these in the text.

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